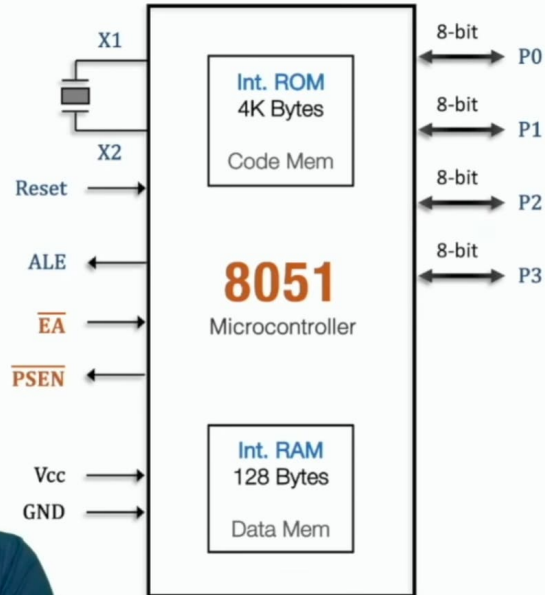


8051

Pin Diagram



8051 | 40 Pins

X1, X2: Clock

Reset: Reset

ALE: Address Latch Enable

EA: Enable External Access

PSEN: Prog Status Enable

VCC, GND: Power Supply

P0: 8-bit I/O Port

P1: 8-bit I/O Port

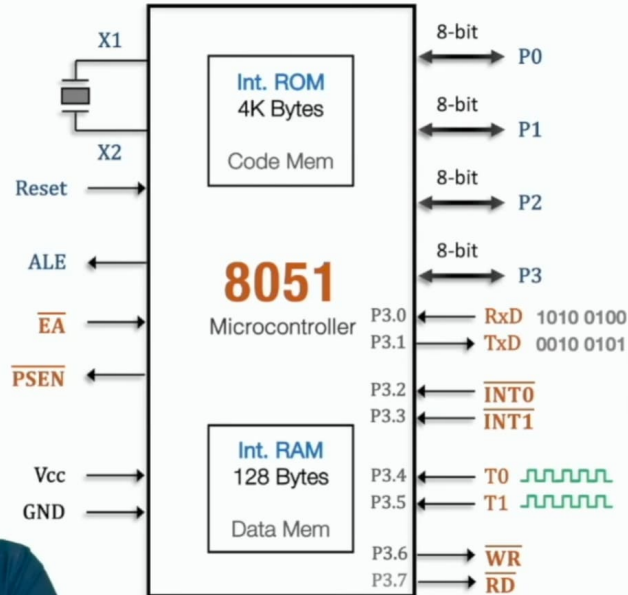
P2: 8-bit I/O Port

P3: 8-bit I/O Port



8051

Pin Diagram



8051 | 40 Pins

X1, X2: Clock

Reset: Reset

ALE: Address Latch Enable

EA: Enable External Access

PSEN: Prog Status Enable

VCC, GND: Power Supply

P0: 8-bit I/O Port

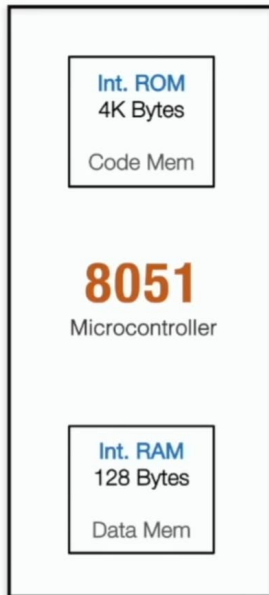
P1: 8-bit I/O Port

P2: 8-bit I/O Port

P3: 8-bit I/O Port

8051

Pin Diagram



8051 | Internal ROM

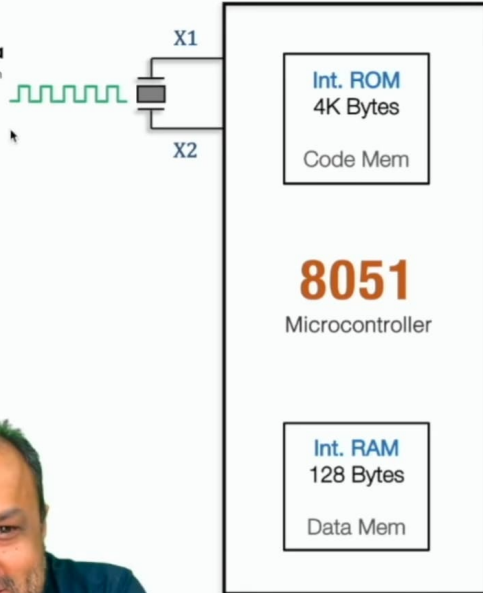
Size: 4 KB

Mainly stores: Programs

8051 | Internal RAM

Size: 128 Bytes

Mainly stores: Data



8051

Pin Diagram



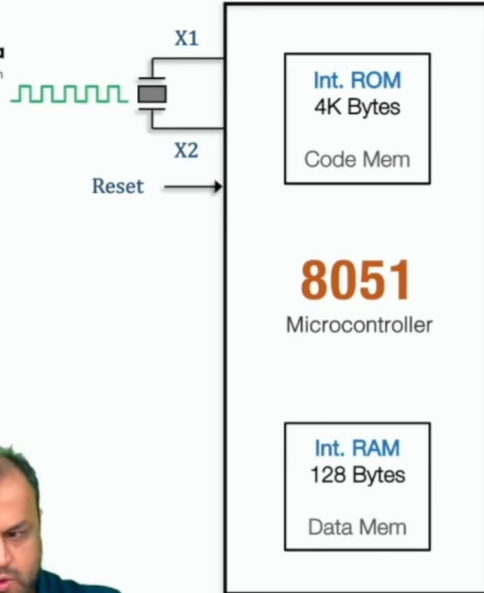
X1, X2 | Clock



Std Crystal Freq: 12 MHz

8051

Pin Diagram



Reset | System Reset

**Processor,
RAM,
I/O ports: Reset**

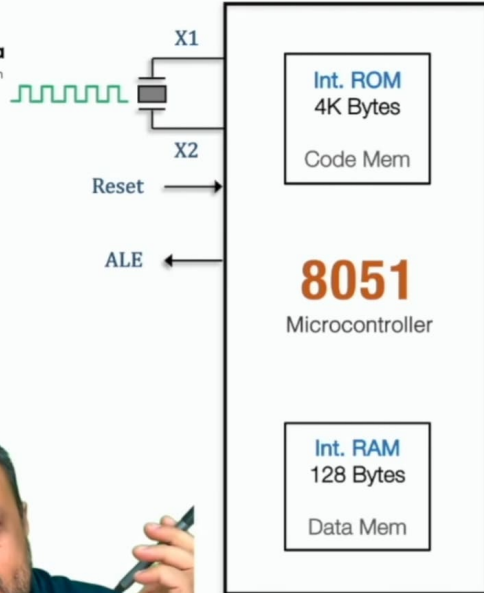
ROM: Contents are **retained**

PC: 0000H
Program starts execution
from location **0000H**.



8051

Pin Diagram



ALE | Address Latch Enable

Address: 16-bits, **A0-A15**
(A0- A7 and A8-A15)

Data: 8-bits, **D0-D7**

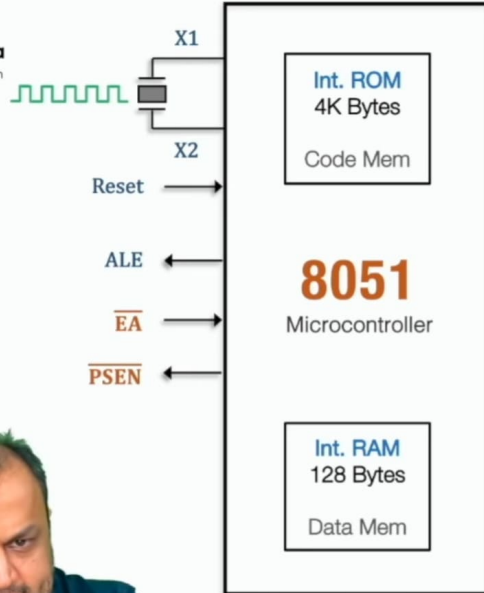
Multiplexing Logic:
A0-A7 and **D0-D7**
Becomes... **AD0 – AD7**

When ALE = 1
AD0 – AD7 carries **A0-A7**

When ALE = 0
AD0 – AD7 carries **D0-D7**

A8-A15: Always address

8051 Pin Diagram



EA | Enable Ext. Access

EA = 0:

Ext. ROM starts from **0000H**

EA = 1:

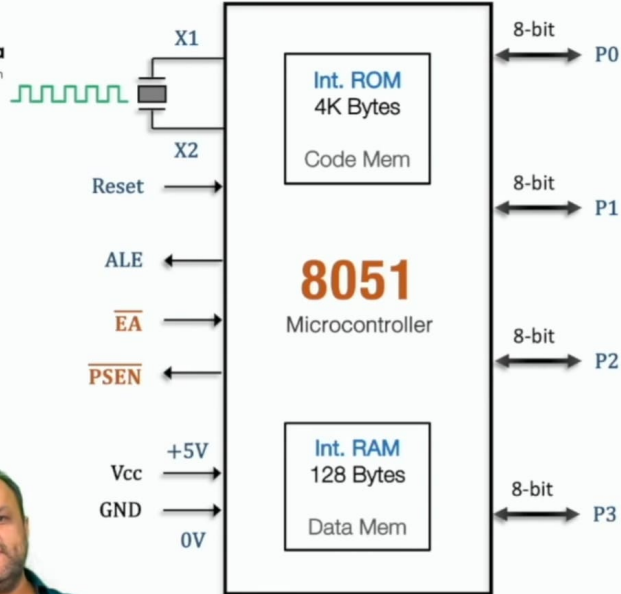
Ext. ROM starts from **1000H**

PSEN | Prog. Status Enable

For Ext ROM:

PSEN acts as **Read** signal

8051 Pin Diagram



I/O Ports | 4, 8-bit, I/O Ports

All Bidirectional

Can be input or output ports

Byte Operations

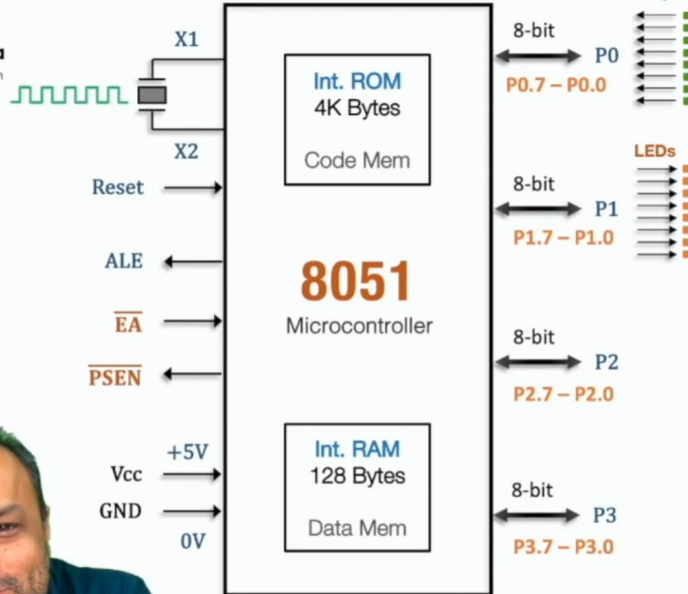
MOV P0, #25H;

Send 25H on P0

00100101 goes on P0



8051 Pin Diagram



I/O Ports | 4, 8-bit, I/O Ports

All Bidirectional

Can be input or output ports

Byte Operations

MOV P0, #25H;

Send 25H on P0

00100101 goes on P0

Bit Operations

SETB P0.0;

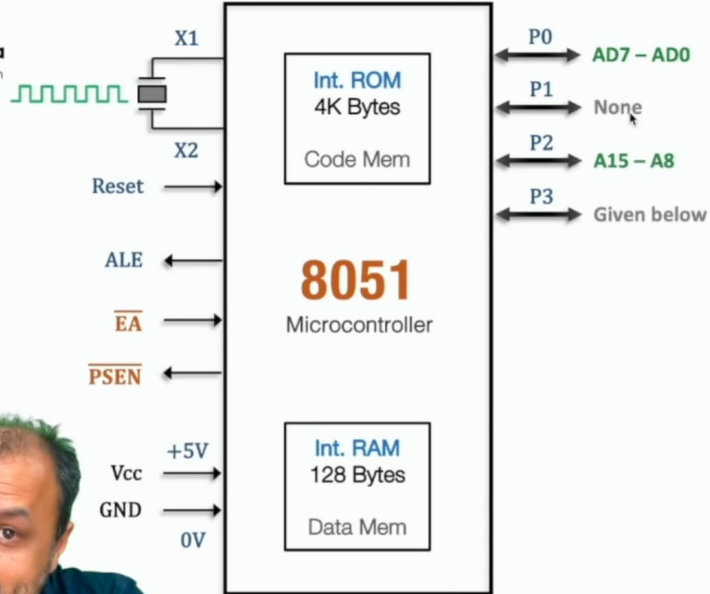
P0.0 becomes 1

CLR P0.0;

P0.0 becomes 0



8051 Pin Diagram



Alternate Functions

P0: AD7 – AD0

Multiplexed address data bus

P1: No alternate function

Always used just as a port

P2: A15 – A8

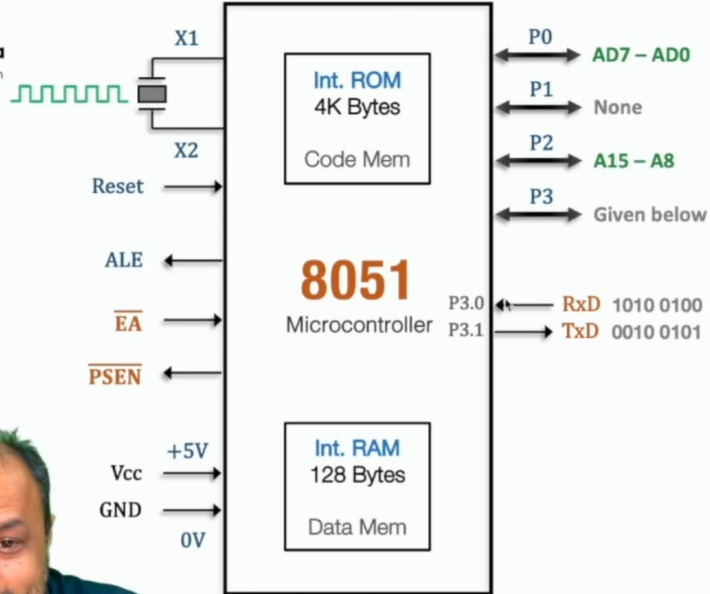
Higher order address bus

P3: Many imp. functions

Powerful features of 8051



8051 Pin Diagram



Alternate Functions of Port3

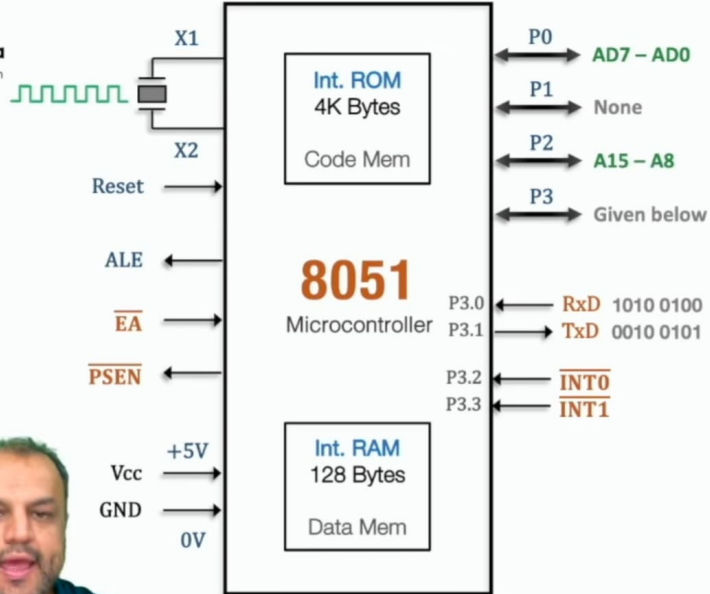
P3.0: RxD

P3.1: TxD

Serial port of 8051



8051 Pin Diagram



Alternate Functions of Port3

P3.0: RxD

P3.1: TxD

Serial port of 8051

P3.2: INT0

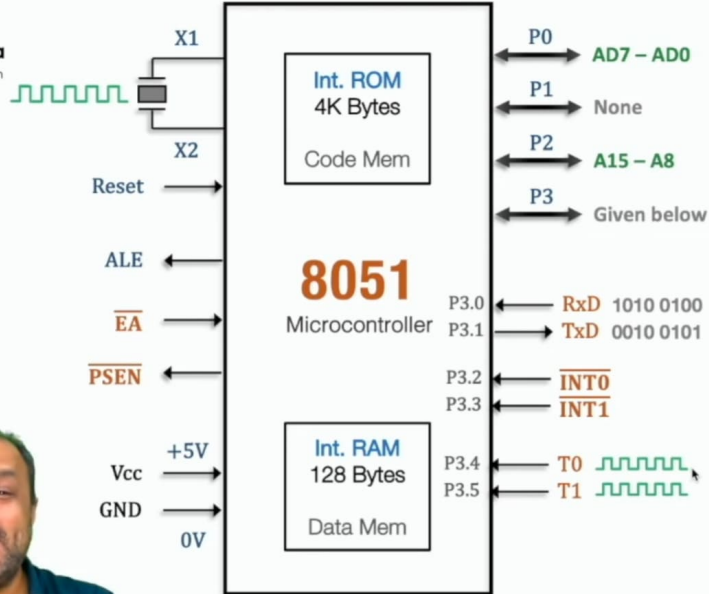
P3.3: INT1

External Hardware Interrupts



8051

Pin Diagram



Alternate Functions of Port3

P3.0: RxD

P3.1: TxD

Serial port of 8051

P3.2: INT0

P3.3: INT1

External Hardware Interrupts

P3.4: T0

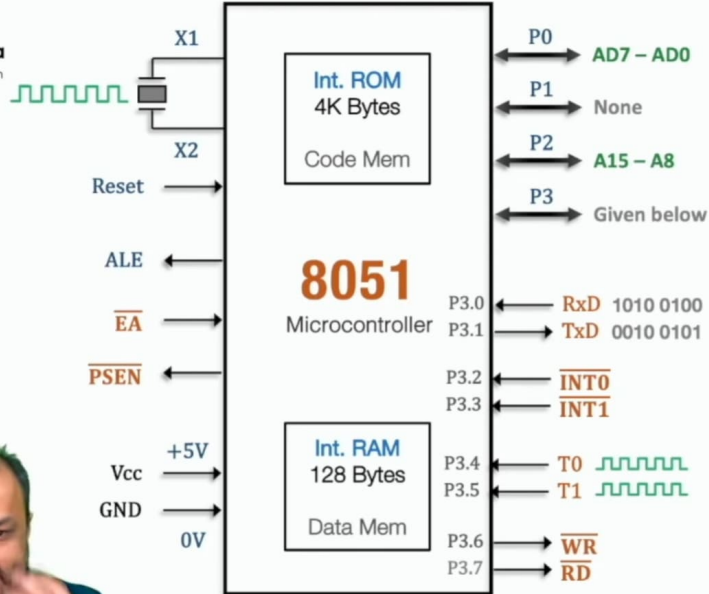
P3.5: T1

Timer clock inputs



8051

Pin Diagram



Alternate Functions of Port3

P3.0: RxD

P3.1: TxD

Serial port of 8051

P3.2: $\overline{INT0}$

P3.3: $\overline{INT1}$

External Hardware Interrupts

P3.4: T0

P3.5: T1

Timer clock inputs

P3.6: $\overline{WR}$

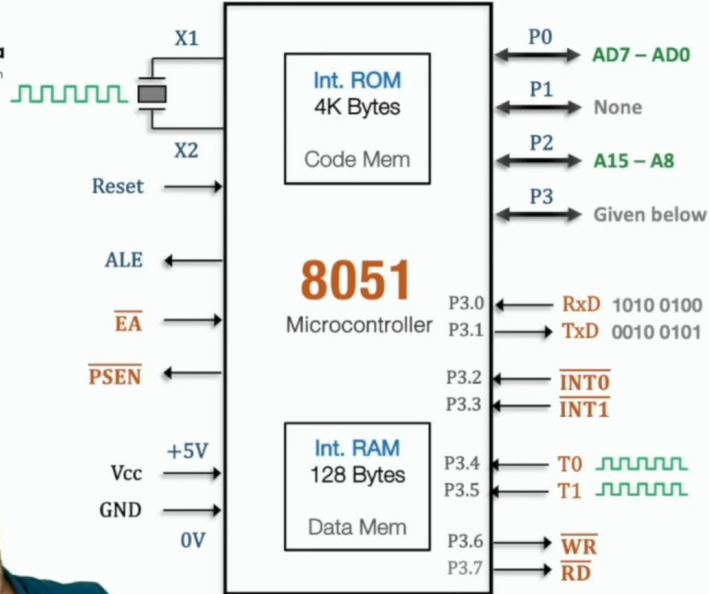
P3.7: RD

Control signals for Ext. RAM



8051

Pin Diagram



External Memory for 8051

Address Bus: 16 - bit

Memory Capacity: 64 KB

Address: 0000H - FFFFH

External ROM

Max Size: 64 KB

Address: 0000H - FFFFH

Read Signal: $\overline{PSEN}$

External RAM

Max Size: 64 KB

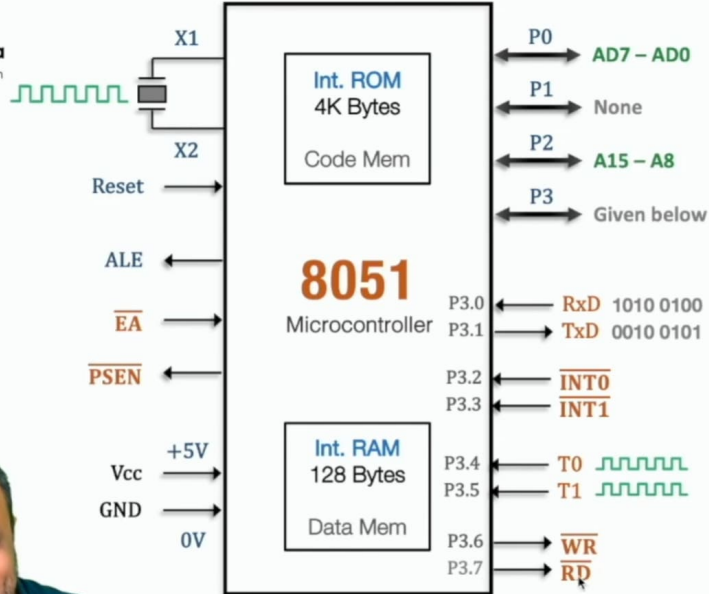
Address: 0000H - FFFFH

Read Signal: $\overline{RD}$

Write Signal: $\overline{WR}$

8051

Pin Diagram



External Memory for 8051

Address Bus: 16 - bit

Memory Capacity: 64 KB

Address: 0000H - FFFFH

External ROM

Max Size: 64 KB

Address: 0000H - FFFFH

Read Signal: $\overline{PSEN}$

External RAM

Max Size: 64 KB

Address: 0000H - FFFFH

Read Signal: $\overline{RD}$

Write Signal: $\overline{WR}$

P0, P2: Address & Data

P3: Control Signals